

An Observation Contract for Public LLMs: Typed Refusal before Crypto Market Decisions

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Abstract

Claim. Public LLMs fail on crypto markets for want of a reliable *observation interface*, not for want of another trading strategy. We contribute a typed observation contract—an anonymized *world-model runtime* (state compiler + hard refuse flag, not a generative simulator)—that turns asynchronous multi-band evidence into a point-in-time (PIT) world bundle. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\text{AI}}$; completeness, honesty (B, U, H) , and WMI/ACWMI expose machine-readable refusal; evidence IDs bind actions to disclosed fields. **Evidence.** On identical OOS asset-days, four live public LLMs (temperature 0) form a four-arm ranking: Blind refuses; Raw over-trades and loses; Ungated abstains only partially (≈ 0.68); Compiled with a hard flag abstains on every thin day (1.0). Full-OOS and band-thick splits reproduce the ranking. This panel never exceeds the production refuse threshold ($\max \text{WMI} \approx 0.093 < 0.2$), so Compiled’s full refusal is a scarce-support stress test. On the open slice ($\text{WMI} \geq 0.05$; $n=1224$), live Ungated CE is positive for every vendor (mean $\approx +0.29$; bootstrap CIs include 0) and a no-LLM rule beats momentum ($\Delta \text{CE}=0.315$, $p=0.034$)—a content lower bound, not Compiled-open trading. Grounding recovers ready/missing/tilt at ≈ 0.97 (Compiled) vs. ≤ 0.23 (Raw). **Falsifiable claim.** Hard gates enforce thin abstention; soft text does not; open-slice fields are nonempty under disclosure and a no-LLM probe.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

observation contract, typed abstention, public LLMs, cryptocurrency, point-in-time, trustworthy AI, world models, systems

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1 Introduction

Core claim (read this first). Public LLMs fail on crypto *before* strategy: they lack a complete, honest, auditable *market observation interface*. Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [13, 15], while generative world models [7, 8, 11] and tool-using finance agents [22–24] typically *assume* a usable observation stream. This paper builds the missing interface—a typed observation contract with a hard refuse flag—not a new trading agent and not a generative simulator.

In one line: *compile asynchronous evidence into a PIT world bundle; refuse when the world is thin; show that live LLMs obey the hard contract and fail without it; treat open-slice content as a lower bound, not a Compiled-open trading claim.*

What we build. We implement the contract as an anonymized market *world-model runtime*: a state compiler from raw multi-band evidence to a PIT-safe bundle for GPT /DeepSeek /GLM /Gemini-class models under a typed refuse boolean. “World model” here means *observation compiler + abstention runtime* (Section 3), not a next-state dynamics model. The compiled world must be **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence).

How we evaluate. **RQ1 (primary):** Can the hard contract *enforce* thin-world abstention across live public LLMs? Four arms on identical days—Compiled (hard flag), Ungated (same content, no flag), Raw (momentum fragment), Blind (no feed). **RQ2:** Does the compiled bundle make analyst answers (ready /missing /tilt) verifiable? **RQ3 (secondary):** Are the vintaged tilts economically nonempty under a transparent no-LLM rule vs. momentum /buy-and-hold?

Contributions. (1) **Interface semantics:** epistemic observations, Π_t , lag reconstruction, completeness/honesty, WMI/ACWMI abstention, evidence-bound actions—enough to define the contract, not a pricing theorem. (2) **Runtime:** vintage collectors, BandPIT previous-close clock, quality-tagged bundles, shocks O_t , multi-vendor adapter, frozen Compiled /Ungated /Raw /Blind protocol. (3) **Live validation:** Blind refuses; Raw over-trades and loses; Ungated ≈ 0.68 on thin mixes; Compiled thin-abstain 1.0 at the production gate; open-slice Ungated CE is positive for all vendors as a content lower bound. Grounding 0.97 vs. Raw 0.04–0.23. The estimand is the four-arm ranking, not ΔCE vs. a strongest-agent baseline.

2 Related Work

World models in AI. World models learn compact states and often dynamics [7, 8, 11]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [14], open financial LLMs [21], memory-augmented trading agents [12, 23], and tool-augmented agents [22, 24] improve agent skill and orchestration. Recent ICAIF work targets adjacent layers: FinAgentBench evaluates agentic retrieval for financial QA [1]; FinSearch builds temporal-aware search agents over market and news feeds [18]; FinResearchBench judges long-horizon research agents with logic-tree protocols [19]; FactorMAD uses multi-agent debate for interpretable alpha mining [3]. Those systems improve how agents fetch, debate, and are scored, but still assume a usable observation stream. We

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study the missing prior step—a PIT-safe, quality-tagged *observation contract* with typed abstention—and evaluate within-model Compiled / Ungated / Raw / Blind under a frozen schema rather than proposing another memory router, retrieval planner, or debate society.

Selective prediction and trustworthy AI. Abstention can be optimal under noise [4, 5]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [13, 15] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [9, 17, 20]. Asset-pricing ML [6, 10, 16] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Observation-Contract Semantics

This section defines the observation contract formally (interface semantics, not a pricing theorem). Readers who want the systems object first may skim to Section 4; the empirical estimand is the four-arm protocol in Section 5.

Definition 3.1 (Epistemic observation). An epistemic observation is

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}) :$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 3.2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Definition 3.3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (should_ai_abstain) and thin_world as first-class fields.

Assumption 3.4 (Bounded lag and increments). Band j has max observation lag Δ_j ; latent state increments satisfy $\|S_u - S_{u-1}\| \leq \bar{\delta}$.

PROPOSITION 3.5 (LAG RECONSTRUCTION BOUND). *Under Assumption 1,*

$$\|\tilde{S}_t - S_t\| \leq C_{\text{del}}\bar{\delta} + C_{\text{noi}}\epsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.5 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

PROPOSITION 3.6 (COMPILATION $\neq$ FEATURE DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

PROPOSITION 3.7 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t , WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

PROPOSITION 3.8 (LOBO CONTENT VS. GATING). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Implication for the runtime. Props. 3.6–3.7 motivate exporting gates and abstain flags rather than tilts alone. Prop. 3.8 motivates the economic check: if deleting a durable band does not change actions, the bundle is empty of content; the probe shows that is not the case here.

REMARK 3.9 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler + abstention runtime.*

4 Runtime Architecture

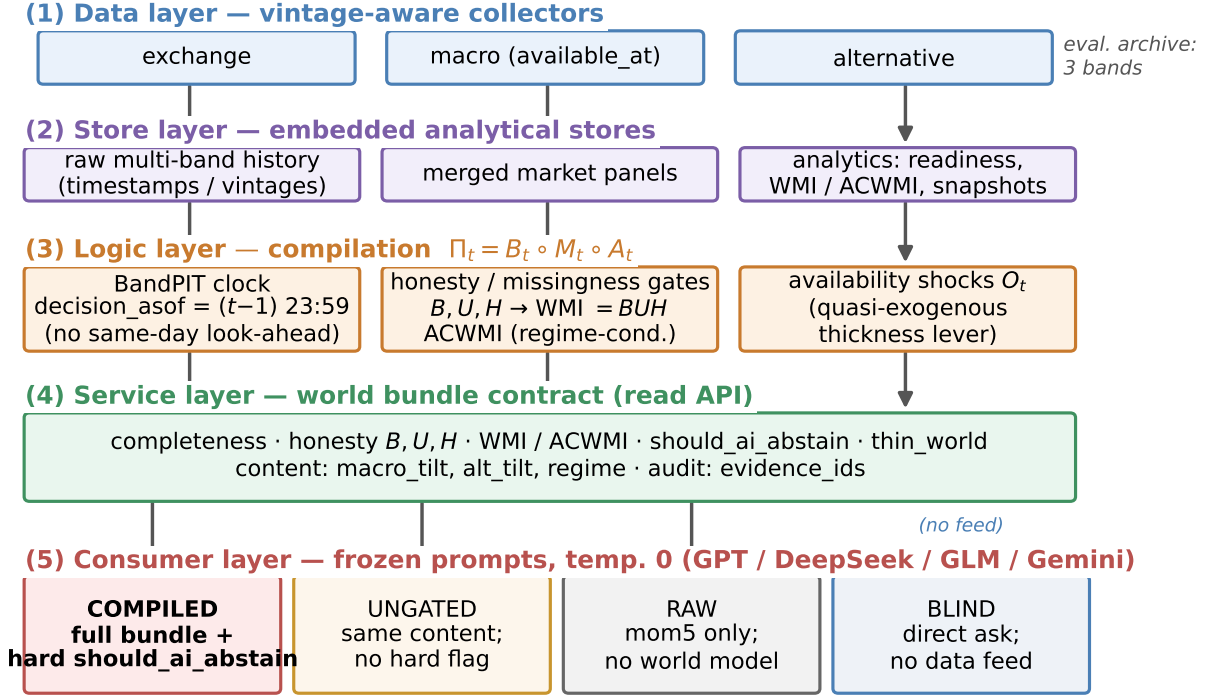
We describe the anonymized prototype as a layered system (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

- (1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro available_at vintages.
- (2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI / ACWMI, paper world snapshots).
- (3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty and missingness gates plus band assembly implement Π_t ; world-quality indices and availability shocks O_t are first-class.
- (4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
- (5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs:* the evaluation archive populates three public bands—exchange klines, funding, and open



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Anonymized runtime architecture. Five numbered layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer defines the frozen four-arm protocol (Compiled / Ungated / Raw / Blind). Live abstention outcomes are in Sec. 5; Blind has no feed from the service layer.

interest via REST; daily macro risk indices (equity volatility, dollar index) with available_at vintages; and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing:* for every asset-day, BandPIT selects the latest observation with timestamp $\leq (t-1) 23:59$, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t . *Outputs:* a ~3,990-row PIT panel (10 assets $\times$ 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

4.3 PIT previous-close clock

For calendar day t , the decision information set is reconstructed at $(t-1) 23:59$. The payoff is the same-calendar-day close-to-close return r_t . This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in decision_asof, and every live run pins timing_protocol to the previous-close protocol.

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetches (raw)	Emitted (bundle)
exchange	OHLKV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H , WMI, ACWMI, should_ai_abstain, thin_world, O_t , evidence_ids

4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, should_ai_abstain=true, and evidence_ids

Table 3: World bundle field groups.

Group	Fields
Timing	decision_asof, previous-close protocol
Completeness	n_{ready} /limited/missing, band statuses
Honesty	B, U, H , main-view / stale gates
Quality	WMI, ACWMI, should_ai_abstain, thin_world
Content	macro_tilt, alt_tilt, regime, cascade
Audit	evidence_ids, EAR required

binding actions to disclosed bands/tilts. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only mom5 (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  },
  "world_model_index": {
    "wmi": 0.015,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
  "audit": {
    "evidence_ids": [
      "band:exchange:missing",
      "band:macro:ready"
    ]
  }
}
```

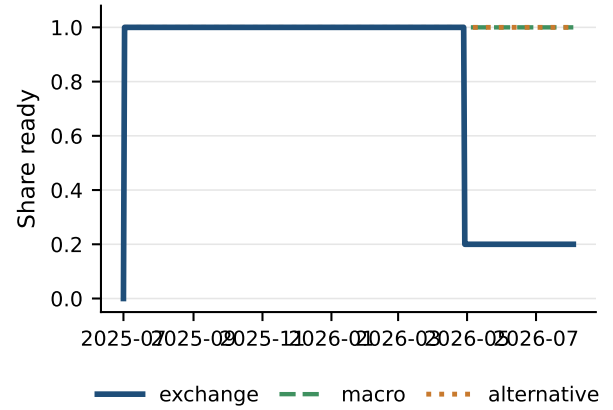
4.5 Four information arms and LLM adapter

- **Compiled (hard contract):** full bundle; frozen prompt *requires* abstain when should_ai_abstain is true.
- **Ungated (disclosure control):** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness (disclosure without enforcement).
- **Raw (thin integration):** mom5 only; no quality index or abstention guidance. This is a common price-only wiring, not an exhaustive tool-agent baseline: the primary estimand is the four-arm ranking (hard vs. soft vs. fragment vs. none), not Δ CE against the strongest possible agent.
- **Blind (direct ask):** “Today is d . How should I trade a ?” with *no* data feed.

Actions $\in \{\text{bullish, bearish, neutral, abstain}\}$. Primary metrics are thin-world abstain rate and EAR proxy; CE is reported as avoided loss under refusal, not as an alpha claim. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive *band-thick* contrast below. The refuse rule sets should_ai_abstain when $\text{WMI} < 0.2$. On this scarce OOS panel max $\text{WMI} \approx 0.093$, so every day is thin and Compiled never opens—a refuse-under-sparse stress test, not a claim the contract never allows action. The adapter uses OpenAI-compatible chat completions; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

**Figure 2: PIT readiness for the three-band evaluation archive (solid / dashed / dotted for grayscale).****Listing 2: Compiled contract (excerpt).**

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why hard refusal is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: when $q(W_t)$ is low, production agents need an enforceable refuse interface (Prop. 3.7). Ungated measures soft judgment from disclosure alone; Raw measures the common price-signal-only integration. The headline result is that the hard contract is reliable across vendors; soft refusal is not.

4.7 Service surface and reproducibility

The service layer exposes read-only objects for the consumer harness: quality-tagged bundles, readiness / WMI–ACWMI snapshots, shock queries O_t , and health metadata. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and gateway URL are configuration, not estimands. Credentials never enter artifacts. Frozen prompts, protocol notes, and table builders for every reported figure/table ship with the submission; an anonymized harness is available to reviewers on request and public on acceptance.

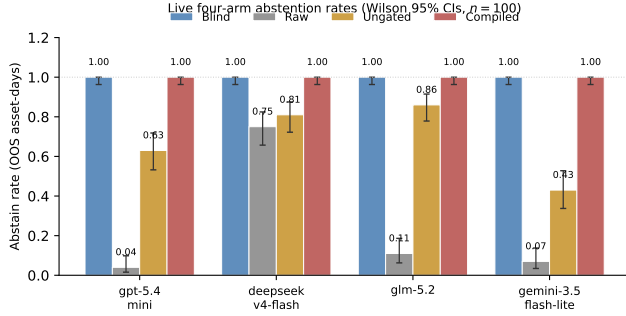
5 Experiments

5.1 Setup

PIT panel ~399 days, 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days ($\approx 2,000$ OOS asset-days). Live RQ1 uses the *same* stratified 100 OOS asset-days across arms (seed fixed),

Table 4: Live abstention rates on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. Δ CE is avoided loss, not an alpha claim.

Model	Thin C	Thin U [CI]	Abs. R [CI]	Δ CE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 3: Live four-arm abstention rates (same 100 days; Wilson 95% CIs). Blind refuses; Raw over-trades; Ungated is vendor-dependent; Compiled enforces the typed refuse contract.**

temperature 0, frozen prompts, and an OpenAI-compatible gateway; a checkpointed full-OOS sweep tests population scale. Vendor models are GPT-, DeepSeek-, GLM-, and Gemini-class flash/mini IDs listed in Table 4. Scarce/thin labels bind on OOS by design; Sec. 5.4 covers fully populated days.

5.2 RQ1 (primary): Live interface behavior across information arms

Table 4 and Figure 3 report the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. This is prompt-contract obedience by design, not spontaneous thinness discovery. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86); DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and gemini-3.5-flash-lite falls to 0.43 and trades. Unconditional Ungated CE (abstain days as cash) spans -1.04 to $+0.02$; restricting to days the model actually acts, three of four vendors post large negative act-conditional CE (mean ≈ -1.9), so soft judgment does not safely convert disclosure into profitable action. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Mean Δ CE (Compiled–Raw) is $+1.07$ and positive for every vendor, because Compiled sits out losses that Raw incurs on sparse support.

Full-OOS replication. To check that the 100-day ranking is not a subsample artifact, we checkpointed the three data arms on the full OOS panel ($n=2000$ asset-days; every OOS day is thin-world

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Table 6: Full-OOS band-thick / open slice (3/3 ready $\equiv$ WMI ≥ 0.05). Production Compiled abstains 1.0; Ungated CE point estimates are positive (bootstrap CIs include 0); Raw over-acts.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM $n=398$; Ungated GLM $n=932$; others $n=1224$. Open-slice Ungated mean CE $\approx +0.31$ (block bootstrap $B=5,999$ reps; one-sided $p \approx 0.21$ vs. CE ≤ 0).

under the scarce-support design). Compiled thin-abstain is 1.000 [.998, 1.000] for all four vendors. Ungated rates tighten the Wilson intervals, with mean thin-abs. 0.56 (GPT 0.666 [.645, .686]; GLM 0.679 [.654, .703] on $n=1402$ completed calls; DeepSeek 0.540 [.518, .561]; Gemini 0.359 [.338, .380]). Vendor ranking within Ungated shifts relative to the 100-day sample (DeepSeek is less cautious at scale), but no vendor matches Compiled. Where Raw is trusted, over-trading remains sharp: GPT and Gemini abstain at 0.000 [.000, .002]; DeepSeek stays cautious at 0.748 [.728, .767] ($n=1925$). Because the full OOS window is entirely thin, Compiled CE is identically zero by construction; thick-day economics are deferred to Sec. 5.4.

Blind arm. On the same 100 asset-days, the no-feed prompt “Today is d . How should I trade BTC?” yields abstention on 100% of days for all four models (CE = 0; zero directional actions). Without a feed, the models refuse. The costly failure mode is the middle case: the same model given a Raw fragment starts trading (abstain as low as 0.04) and loses. Compiled matches Blind’s refusal on thin days by contract, while still exposing the structured world that RQ2 and RQ3 show is usable.

Band-thick / open-slice days. Archive readiness alone is not a refuse policy. On full-OOS *band-thick* days (3/3 ready; 61% of OOS, $n \approx 1224$; $\equiv$ WMI ≥ 0.05), production Compiled still abstains at 1.0 (WMI < 0.2). Ungated on the same days keeps compiled content without the hard boolean: mean abs. ≈ 0.48 , CE positive for every vendor (mean $\approx +0.29$; Table 6). Block bootstrap ($B=5,999$ reps) CIs include zero (mean CE $\approx +0.31$, one-sided $p \approx 0.21$); we report a content lower bound, not a trading edge, and *not* Compiled-open (the frozen prompt still forces abstain when the world is judged thin even if the boolean were false). Raw remains risky: GPT and Gemini abstain at 0.0 with CE -0.28 and -0.25 .

Table 7: Grounding workflow (50 days): ready / missing / tilt signs.

Model	Ready F1		Missing F1		Tilt-sign acc.	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Inside the Ungated arm. Ungated rationales cite disclosed world fields on 53–93% of days, so models *read* the world; they differ in acting on it. Vendor-specific caution—not missing disclosure—drives the gap the typed contract closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only mocks ignore thin-world flags (thin-abs. 0). Offline results are protocol checks; live vendors are the main claim.

When would the contract open? Compiled never trades under max WMI $\approx 0.093 < 0.2$ —a scarce-support stress test, not a permanently closed contract. A WMI gate of 0.05 would clear the boolean on $n=1224$ open days (ACWMI already ≥ 0.25), but would *not* clear the Compiled prompt’s separate “world is thin” clause. Live evidence on that slice is Ungated content (Table 6) plus the no-LLM rule vs. momentum ($\Delta\text{CE}=0.315$, $p=0.034$; Table 9). Denser vintages can raise the WMI floor without changing contract semantics.

5.3 RQ2: Non-trading cognition—grounding workflow

The compiled world should also support analysis, not only trade/abstain decisions. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report $\text{sign}(\text{macro_tilt})$ and $\text{sign}(\text{alt_tilt})$, and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 7: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency judgments alone are weak once completeness is scoped to real archive bands—models may call a 3/3-ready day “sufficient” while WMI still marks the world thin. The sharper claim is verifiability: only the compiled world lets the model state what is ready, missing, and signed, in a form that can be checked. This is the analyst-facing counterpart of Prop. 3.5’s missingness term.

5.4 RQ3: Secondary content probe

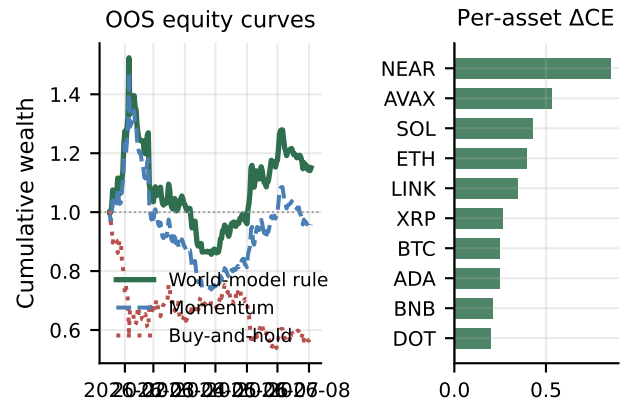
RQ1–RQ2 concern the interface. RQ3 asks whether the compiled tilts are economically nonempty. A transparent rule shares return inputs with momentum and adds vintaged $\text{macro_tilt} / \text{alt_tilt}$ —no LLM in the loop. Out-of-sample (199 days $\times$ 10 assets) this content rule beats the baselines (Table 8, Figure 4): Sharpe 0.764 / CE +0.130 versus momentum (0.096, -0.206) and buy-and-hold (-1.404 ,

Table 8: Secondary content probe: world-model rule vs. traditional baselines (no LLM).

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	0.764	+0.130
Mechanism – Momentum	—	0.334 ($p=0.034$)

Table 9: Open/dense split of the content probe (OOS). Open = WMI $\geq 0.05 \equiv$ 3/3 bands ready.

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.336
Open / thick ($n=1224$)	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

**Figure 4: Secondary content probe (no LLM). Left: equity curves; right: ΔCE vs. momentum positive for every asset.**

−1.163). The pre-specified contrast mechanism – momentum is significant under block bootstrap ($\Delta\text{CE}=0.334$, $p=0.034$). ΔCE is positive for all ten assets (Figure 4, right). LOBO (Prop. 3.8) attributes the gain to band content (deleting macro/alternative collapses to momentum); gaps survive 10–25bps costs. A 2017–2026 audit with non-vintaged proxy tilts shows no edge ($\Delta\text{CE} \approx 0$, $p=0.81$): value appears where the vintage-compiled world exists. The result supports nonempty tilts, not an LLM trading ranking.

Within-archive dense / open world. Among the three archive bands, 61% of OOS asset-days are band-thick (3/3 ready) and 39% are band-thin (exchange gap; Figure 2). On this panel, band-thick coincides with counterfactual open days ($\text{WMI} \geq 0.05$). Table 9: on that open/thick slice the content rule acts and beats momentum ($\Delta\text{CE}=0.315$; CE +0.061 vs. -0.254), stronger than on exchange-gap days (0.129)—complementary to RQ1’s refuse-under-thin test.

5.5 Discussion and threats

What the results mean. Three facts carry the paper. (i) *Thin days must refuse*: soft “consider abstaining” text does not; Compiled’s hard gate does, across live vendors (RQ1–RQ2). (ii) *Disclosure without enforcement is unstable on thin mixes*: Ungated abstains only ≈ 0.68 (GLM 0.86 vs. Gemini 0.43) and often destroys CE versus cash when it acts. (iii) *Open-slice content is a nonempty lower bound*: Ungated CE is positive for all four vendors on WMI ≥ 0.05 (Table 6; bootstrap CIs include zero), and RQ3’s no-LLM rule is significant vs. momentum. Raw and Blind show unstructured or empty context cannot replace typed readiness. We claim a typed observation interface—not LLM trading SOTA, not a generative simulator, not a full causal theory of crypto markets.

Relative to FinMem /FinAgent [23, 24] and ICAIF retrieval/debate/judge lines [1, 3, 18, 19], we contribute an observation compiler and refuse interface, not a new memory/search/debate router. Relative to return-prediction LLM studies [14], we score refusal and auditability before PnL.

Design rationale. Three choices are deliberate. (i) Export gates, not only features: completeness, honesty, and should_ai_abstain make refusal machine-readable rather than a free-text apology. (ii) Previous-close PIT over wall-clock streaming: every live decision cites a recomputable decision_asof, avoiding same-day look-ahead. (iii) Four arms instead of one Sharpe number: Compiled / Ungated / Raw / Blind separates typed enforcement, soft judgment, thin integrations, and no feed.

Threats to validity. Compiled thin abstention is prompt-contract obedience by design; Ungated removes the boolean on purpose. The archive is scarce (max WMI ≈ 0.093), so production Compiled never opens at 0.2. Open-slice Ungated is a content lower bound, not Compiled-open trading: the prompt’s “world is thin” clause can still force abstain if the boolean were cleared, and CE bootstrap CIs include zero. Mixed-sample act-conditional Ungated CE is negative for three of four vendors; open-slice point estimates are positive—regime matters. Vendors are flash/mini at temperature 0; GLM Raw full-OOS is incomplete; denser bands are absent. Raw is common mom5 wiring for the four-arm ladder, not dominance over tool-using agents.

Scope. We do not claim LLM trading SOTA or generative dynamics. Natural extensions include richer bands and thick-regime live trading once WMI leaves the scarce floor. Camera-ready will restore author identity and release the anonymized harness deferred during review.

6 Conclusion

Public LLMs need a market they can safely observe before they decide. We contribute a typed observation contract—an anonymized state-compiler runtime with a hard refuse flag—for asynchronous crypto feeds. The live ranking is the point: Blind refuses, Raw overtrades and loses, Ungated is unstable on thin mixes, and Compiled enforces thin-world abstain 1.0 at the production gate; open-slice Ungated and a no-LLM content probe give a nonempty lower bound, not a Compiled-open trading claim. Grounding recovers ready/missing/tilt answers at ≈ 0.97 vs. Raw ≤ 0.23 . Hard gates enforce thin

refusal; soft text does not; the contribution is the interface, not a trading leaderboard.

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